

TRACKSERV FRONTEND DEVELOPMENT REPORT

Step 1: Development of the Login, Registration and Role Selection Interface

The first part of the frontend development was the creation of the login and registration system. The login interface was designed to provide a clear and simple way for users to access TrackServ.

The login page contains the required fields for users to enter their login information, including their email/username and password. A login button was also included to allow the user to submit their details.

A registration section was developed for new users. This allows people who do not already have an account to create one and provide the necessary information to use TrackServ.

The frontend also considered the different types of users within the system. Normal users and administrators require different levels of access, so the interface was designed to support separate user and administrator login access.

How this helps TrackServ

The login and registration functionality is important because it provides the entry point into the TrackServ system. It allows the system to identify users and control what they can access.

For example:

User → Login → Authentication → User Dashboard

An administrator follows a separate access path, allowing them to access administrative functions.

This improves the security and organisation of TrackServ because normal users do not need access to administrative functions, while administrators can access the tools required to manage reports, users, assignments and locations.

Step 2: Development of the Landing/Home Page

The next part of the frontend was the development of the TrackServ landing page.

The landing page was designed as the first page that a visitor can see when accessing the system. It introduces TrackServ and explains its purpose, which is to help citizens report municipal issues and monitor the progress of those issues.

The page includes:

- * TrackServ branding.
- * Navigation options.
- * A main introduction section.
- * Information explaining the purpose of the platform.
- * A Report Issue button.
- * A Track Status button.
- * Statistics showing information about reports.
- * A map/location section.
- * Information about community participation.
- * Calls to action encouraging users to participate.

The page was designed to be simple enough for a first-time visitor to understand what TrackServ does without needing extensive instructions.

How this helps TrackServ

The landing page acts as the main entry point to the system.

A citizen can immediately understand that TrackServ allows them to:

1. Report a municipal problem.
2. Track the progress of their report.
3. Interact with the community.
4. Access other services provided by the platform.

This makes the system easier to use and encourages citizens to actively report problems in their communities.

Step 3: Development of the User Profile Page

A user profile interface was also developed.

The profile page provides users with a central location where they can view information about their account and activity on TrackServ.

The design contains areas for:

- * User profile information.
- * User statistics.
- * Report activity.
- * Completed activities.
- * Badges/achievements.
- * Community information.
- * User-related settings.
- * Additional profile information.

The profile interface was designed to give the user an overview of their participation on the platform.

How this helps TrackServ

The profile page helps users keep track of their own activity.

For example, users can see information related to:

- * How many reports they have submitted.
- * The status of their activities.
- * Their contribution to the community.
- * Their achievements.

This encourages users to remain active on TrackServ because they can see their contribution and progress.

It also gives TrackServ a more personalised experience rather than making the platform feel like a basic reporting form.

Step 4: Development of the About Page

The About page was developed to explain the purpose and goals of TrackServ.

The page was designed around the idea of empowering citizens and building better communities.

It contains information explaining:

- * What TrackServ is.
- * What the platform is intended to achieve.
- * How citizens can participate.
- * The importance of reporting municipal problems.
- * The benefits of community involvement.
- * TrackServ's vision.
- * TrackServ's mission.
- * Important statistics relating to the platform.

The page also includes visual elements and sections that make the information easier to understand.

How this helps TrackServ

The About page helps users understand why the system exists and how it benefits the community.

This is important because users are more likely to use a system when they understand its purpose.

It also provides transparency by explaining the role TrackServ plays in connecting citizens with the reporting and management of municipal issues.

Step 5: Development of the Community/Social Media Section

A Community page was developed to give TrackServ a social and interactive element.

Instead of only allowing users to submit reports, the community section allows users to interact with information and with other members of the community.

The interface was designed in a social-media-style format and contains features such as:

- * Community posts.
- * User profiles.
- * Post content.
- * Images/media.
- * Likes or reactions.
- * Comments.
- * Trending topics.
- * Community members.
- * Community statistics.

- * Interaction options.

The design allows information to be presented in a familiar social-media-style layout.

How this helps TrackServ

This feature makes TrackServ more than just a reporting system.

It allows citizens to share information and interact with community-related issues.

For example, if several people are experiencing the same problem in an area, the community section can help create awareness around that issue.

This can encourage:

- * Community participation.

- * Communication.

- * Awareness of local problems.

- * Collaboration between citizens.

- * Greater engagement with TrackServ.

The community section therefore supports the social side of the TrackServ system.

Step 6: Development of the Report an Issue Page

One of the most important frontend components was the Report an Issue page.

This page allows citizens to formally submit a municipal problem through TrackServ.

The interface was designed as a structured reporting form.

It contains areas for entering information such as:

- * Report title/details.

- * Issue category.

- * Description of the problem.
- * Location.
- * Supporting information.
- * Map/location information.
- * Submission controls.

The page was designed to make the reporting process straightforward for the user.

The location section is particularly important because municipal problems need to be associated with a specific location.

The map component was also incorporated into the overall design so that location information can be represented visually.

How this helps TrackServ

This page forms one of the core functions of TrackServ.

The basic process is:

Citizen identifies problem → Opens Report Issue → Enters information → Provides location → Submits report → Report enters TrackServ system

Once connected to the backend and database, the information submitted through this page can be stored and used by administrators and staff.

This means the frontend provides the actual interface through which a real-world municipal problem becomes a digital report inside TrackServ.

Step 7: Development of the View Reports Page

The View Reports page was developed to allow users to view and monitor submitted reports.

The page contains a dashboard-style layout with information such as:

- * Total reports.
- * Reports in different statuses.
- * Report titles.
- * Report images.
- * Categories.
- * Dates.
- * Status information.
- * Report-related actions.

Different report statuses can be visually distinguished, making it easier for users to understand what is happening with their reports.

For example, a report can move through stages such as:

Submitted → Under Review → In Progress → Resolved

How this helps TrackServ

This feature provides transparency.

Instead of submitting a complaint and having no idea what happens afterwards, the user can return to TrackServ and check its progress.

This improves trust in the system because citizens can see that their reports are being processed.

It also reduces the need for users to repeatedly contact municipal offices simply to ask about the status of a report.

Step 8: Development of the Main User Dashboard

The frontend also included a user dashboard that brings important information together in one place.

The dashboard provides users with an overview of their activity and relevant TrackServ information.

The interface includes dashboard cards, report information, maps, statistics and other important information.

The dashboard is designed to give users a quick overview without requiring them to open every individual page.

How this helps TrackServ

The dashboard improves usability because users can quickly see important information in one location.

For example, users can quickly identify:

- * Their recent reports.
- * Report statuses.
- * Location information.
- * Activity updates.
- * Relevant statistics.

This makes TrackServ faster and easier to navigate.

Step 9: Development of the Admin Assignments Component

After the main user-facing pages were developed, the frontend work expanded into the administrator side of TrackServ.

An admin Assignments component was developed.

The interface allows administrators to view and manage assignments relating to reported municipal problems.

The design contains:

- * Total assignments.
- * Assignments in progress.
- * Completed assignments.
- * Overdue assignments.
- * Assignments due during the week.
- * Search functionality.
- * Assignee filters.
- * Status filters.
- * Priority filters.
- * Date filters.
- * Assignment tables.
- * Assignment IDs.
- * Report information.
- * Assigned staff members.
- * Teams.
- * Priorities.
- * Assignment dates.
- * Due dates.
- * Assignment actions.
- * Assignment summary charts.

- * Priority breakdown.

- * Top assignees.

How this helps TrackServ

This component is important for the operational side of TrackServ.

Once a citizen submits a problem, administrators need to make sure that the problem is actually dealt with.

The assignment system allows administrators to allocate reports to appropriate staff or teams.

For example:

Citizen Report → Administrator Reviews Report → Assignment Created → Staff Member/Team Assigned → Work Completed

This helps TrackServ ensure that reports do not simply remain in the database without action.

It also makes it easier for administrators to monitor overdue and incomplete work.

Step 10: Development of the Admin Locations Component

An Admin Locations page was also developed.

This page focuses on the geographical information associated with TrackServ reports.

The interface contains:

- * Total locations.

- * Active reports.

- * Reports in progress.

- * Resolved reports.

- * Overdue reports.

- * Interactive map area.
- * Map and satellite options.
- * Map markers.
- * Location filters.
- * Location overview.
- * Top areas.
- * Recent locations.
- * Report titles.
- * Addresses/locations.
- * Categories.
- * Statuses.
- * Reported-by information.
- * Dates.
- * Location actions.
- * Category filters.
- * Status filters.
- * Date filters.

The page provides both a geographical and tabular view of reported issues.

How this helps TrackServ

Location information is extremely important for municipal issue management.

The map allows administrators to see where problems are occurring.

For example, if many reports are concentrated in one area, administrators can identify that area as requiring attention.

The location component therefore helps with:

- * Identifying problem areas.
- * Prioritising municipal work.
- * Monitoring reports geographically.
- * Finding patterns.
- * Managing resources.
- * Understanding where issues are occurring.

It also makes it easier for administrators to investigate reports based on their physical location.

Step 11: Development of Navigation Components

The frontend also included common navigation components that could be used throughout TrackServ.

These include elements such as:

- * Sidebar navigation.
- * Top navigation.
- * Search bar.
- * User/account area.
- * Notifications.
- * Navigation links.
- * Footer.

The sidebar provides access to different sections of the system, while the top navigation gives users access to commonly used functions.

How this helps TrackServ

Without consistent navigation, users would struggle to move between the different sections of the system.

The navigation structure allows users to move between areas such as:

Dashboard → Reports → Community → Profile → Other Features

On the administrator side, navigation provides access to:

Dashboard → Reports → Assignments → Users → Categories → Locations → Analytics → Settings

This makes the system easier to understand and reduces unnecessary navigation.

Step 12: Development of the Search Functionality

A search bar was incorporated into the TrackServ interface.

The search functionality is intended to make it easier to locate relevant information such as:

- * Reports.

- * Users.

- * Assignments.

- * Locations.

- * Other system information.

How this helps TrackServ

As the amount of information in TrackServ increases, manually looking through every report or assignment becomes inefficient.

The search function allows users and administrators to locate information more quickly.

For administrators, this can be particularly useful when dealing with a large number of reports and assignments.

Step 13: Development of the Footer

A common footer component was developed for the relevant pages.

The footer provides additional information and navigation/support-related content at the bottom of the interface.

Using a reusable footer means that the same design can be applied consistently across different pages.

How this helps TrackServ

A consistent footer makes the application look more professional and provides users with additional navigation or support information.

It also improves consistency because users see familiar interface elements throughout TrackServ.

Step 14: Use of React for the Frontend

The frontend was developed using React as the main framework.

React was used because TrackServ is an interactive system that contains multiple pages and reusable interface elements.

The frontend was structured into components rather than treating the entire system as one large page.

Examples of components include:

- * Login component.
- * Registration component.
- * Navigation component.
- * Sidebar.
- * Footer.
- * Dashboard cards.
- * Report forms.
- * Report tables.
- * Community posts.
- * Map sections.
- * Admin components.

How this helps TrackServ

Using reusable React components makes the system easier to maintain.

If a component needs to be changed, the change can be applied to the relevant component rather than rebuilding every page from scratch.

React also supports interactive interfaces, which is important because TrackServ requires users to:

- * Submit information.
- * View reports.
- * Filter information.
- * Navigate between pages.
- * Interact with community content.
- * Monitor report statuses.

Step 15: Styling and Responsive Interface Development

CSS were used to structure and style the frontend.

The styling work covered:

- * Page layouts.
- * Colours.
- * Typography.
- * Buttons.
- * Forms.
- * Cards.
- * Tables.
- * Spacing.
- * Icons.
- * Navigation.
- * Responsive layouts.

The interface was designed to remain organised across different screen sizes.

How this helps TrackServ

A municipal reporting system needs to be accessible to different users, including people accessing it from phones, tablets and computers.

A responsive interface helps ensure that important functionality remains usable regardless of the device being used.

This is particularly important because citizens may report an issue while they are physically at the location of the problem using a mobile device.

Step 16: Use of Icons and Data Visualisation

The frontend incorporated libraries such as lucide-react for icons and Recharts for charts and visual representations of information.

Icons were used to make actions and sections easier to recognise.

Charts were used on administrative dashboards to display information such as:

- * Assignment status.
- * Report statistics.
- * Priority breakdowns.
- * Location statistics.
- * Other system information.

How this helps TrackServ

Visual information allows administrators to understand large amounts of data more quickly.

For example, instead of reading hundreds of individual records, an administrator can look at a chart and immediately understand how many reports are:

- * In progress.
- * Completed.
- * Overdue.
- * Awaiting verification.

This supports quicker decision-making.

Step 17: Integration of Maps

The map functionality was incorporated into the TrackServ frontend.

The map is particularly important because TrackServ deals with physical municipal problems.

The frontend provides the area where location information and map data can be displayed.

The map functionality allows reports to be connected to geographical locations.

How this helps TrackServ

The map connects the digital report to the real-world location of the problem.

For example:

Pothole Report → Location Captured → Appears on Map → Administrator Identifies Area → Assignment Created

This makes it easier for municipal staff to locate the problem and respond appropriately.

It also allows administrators to identify areas with multiple reported issues.

Step 18: Database and Backend Connection

The frontend was designed to work together with the backend and database.

Information entered through the frontend can be passed to the backend and stored in the database.

For example:

Report Form

↓

Frontend

↓

Backend



Database



Stored Report

The same information can then be retrieved and displayed through other parts of the frontend.

How this helps TrackServ

This creates a complete information flow within the system.

A citizen does not simply fill in a form that disappears. Their information becomes part of the TrackServ database and can then be accessed by the appropriate users and administrators.

This allows TrackServ to function as a complete information system rather than just a website.

Step 19: Authentication Integration

The frontend was designed to work with authentication services such as Supabase Authentication.

The intended process is:

User



React Login Page



Authentication



Successful Login



Dashboard

Authentication ensures that users can access the appropriate part of the TrackServ system.

How this helps TrackServ

Authentication helps protect information and prevents unauthorised users from accessing restricted areas.

This is particularly important for the administrator section because administrators have access to information and functions that ordinary users should not have.

Step 20: Overall Frontend Integration

After all the individual pages and components were completed, they were brought together into one consistent TrackServ frontend.

The different components work together rather than functioning as unrelated pages.

The complete user flow can be represented as:

User visits TrackServ



Landing Page



Register / Login



User Dashboard



Report an Issue



Submit Location and Report Information



Report Stored in System



View Reports



Track Report Status



Community Interaction



User Profile

At the administrative level, the flow becomes:

Report Submitted



Administrator Receives Report



Administrator Reviews Report



Assignment Created



Staff/Team Assigned



Progress Monitored



Location Monitored



Issue Resolved



Status Updated

Step 21: Overall Contribution of the Frontend to TrackServ

The completed frontend provides the user-facing layer of TrackServ.

It allows citizens to interact with the system while also giving administrators the interfaces required to manage municipal reports.

The frontend connects all the major functions of TrackServ, including:

- * User registration.
- * Login and authentication.
- * Reporting municipal problems.
- * Capturing locations.
- * Viewing reports.
- * Tracking report progress.
- * Community interaction.
- * User profiles.
- * Administrative assignments.
- * Location management.
- * Data visualisation.
- * Search and filtering.
- * Navigation.
- * Dashboard monitoring.

Overall, the frontend makes TrackServ functional from the user's point of view. Instead of users having to interact directly with databases or backend systems, they are provided with a clear interface through which they can submit information, view information and track activities.

The frontend also supports administrators by providing dashboards and management interfaces that allow them to monitor reports, assign work, identify problem areas and follow the progress of municipal issues.

Therefore, the completed frontend forms the connection between the citizen, TrackServ system, administrators, municipal staff and the underlying database.

Complete system flow

CITIZEN



TRACKSERV FRONTEND



REPORT / LOCATION / USER INFORMATION



BACKEND & DATABASE



ADMINISTRATIVE SYSTEM



ASSIGNMENT TO STAFF



MUNICIPAL ISSUE ADDRESSED



STATUS UPDATED



CITIZEN CAN TRACK THE PROGRESS